Table 2.19.2: Summary of the recommended correlations for main thermophysical properties of molten bismuth (Bi)

p ~ 0.1 MPa, except the critical parameters and the saturated vapour pressure

Property, parameter	SI unit	Correlation	Temp. range (K)	Estimated error ±
Melting temperature	K	$T_{M,0} = 544.6$	n/a	0.3
Latent heat of melting	kJ kg^{-1}	$Q_{M,0} = 53.3$	n/a	0.3
Boiling temperature	K	$T_{B,O} = 1\,831$	n/a	6
Latent heat of boiling	kJ kg^{-1}	$Q_{B,0} = 856.2$	n/a	1.9
Critical temperature	K	$T_c = 4\,500$	n/a	500
Critical density	kg m^{-3}	$\rho_c = 2\,800$	n/a	200
Critical pressure	MPa	$p_c = 135$	n/a	15
Saturated vapour pressure	Pa	$p_s = 2.67 \times 10^{10} \cdot \exp(-22\,858/T)$	545-1 831	40%
Surface tension	N m^{-1}	$\sigma = (420.8 - 0.081 \cdot T) \times 10^{-3}$	545-1 400	3%
Density	kg m^{-3}	$\rho = 10\,725 - 1.22 \cdot T$	545-1 300	0.5%
Sound velocity	m s^{-1}	$u_s = 1\,616 + 0.187 \cdot T - 2.2 \times 10^{-4} \cdot T^2$	545-1 800	2%
Bulk modulus	Pa	$B_s = (30.09 - 2.441 \times 10^{-3} \cdot T - 3.913 \times 10^{-6} \cdot T^2) \times 10^9$	545-1 300	5%
Isobaric specific heat	J kg^{-1} K^{-1}	$c_p = 118.2 - 5.934 \times 10^{-3} \cdot T + 7.183 \times 10^6 \cdot T^{-2}$	545-1 300	4%
Dynamic viscosity	Pa s	$\eta = 4.456 \times 10^{-4} \cdot \exp(780/T)$	545-1 300	5%
Electric resistivity	Ω m	$r = (98.96 + 0.0554 \cdot T) \times 10^{-8}$	545-1 420	1.5%
Thermal conductivity	W m^{-1} K^{-1}	$\lambda = 7.34 + 9.5 \times 10^{-3} \cdot T$	545-1 000	(15%)

Table 2.19.3: Summary of the recommended correlations for main thermophysical properties of molten LBE

p ~ 0.1 MPa, except the critical parameters and the saturated vapour pressure

Property, parameter	SI unit	Correlation	Temp. range (K)	Estimated error ±
Melting temperature	K	$T_{M,0} = 398$	n/a	1
Latent heat of melting	kJ kg^{-1}	$Q_{M,0} = 38.6$	n/a	0.3
Boiling temperature	K	$T_{B,O} = 1\,927$	n/a	16
Latent heat of boiling	kJ kg^{-1}	$Q_{B,0} = 856$	n/a	5
Critical temperature	K	$T_c = 4\,800$	n/a	500
Critical density	kg m^{-3}	$\rho_c = 2\,200$	n/a	200
Critical pressure	MPa	$p_c = 160$	n/a	70
Saturated vapour pressure	Pa	$p_s = 1.22 \times 10^{10} \cdot \exp(-22\,552/T)$	670-1 927	60%
Surface tension	N m^{-1}	$\sigma = (448.5 - 0.08 \cdot T) \times 10^{-3}$	420-1 400	3%
Density	kg m^{-3}	$\rho = 11\,065 - 1.293 \cdot T$	400-1 300	0.8%
Sound velocity	m s^{-1}	$u_s = 1\,855 - 0.212 \cdot T$	400-1 100	1%
Bulk modulus	Pa	$B_s = (38.02 - 1.296 \times 10^{-2} \cdot T + 1.32 \times 10^{-6} \cdot T^2) \times 10^9$	400-1 100	3%
Isobaric specific heat	J kg^{-1} K^{-1}	$c_p = 164.8 - 3.94 \times 10^{-2} \cdot T + 1.25 \times 10^{-5} \cdot T^2 - 4.56 \times 10^5 \cdot T^{-2}$	400-1 100	7%
Dynamic viscosity	Pa s	$\eta = 4.94 \times 10^{-4} \cdot \exp(754.1/T)$	400-1 200	8%
Electric resistivity	Ω m	$r = (90.9 + 0.048 \cdot T) \times 10^{-8}$	403-1 100	6%
Thermal conductivity	W m^{-1} K^{-1}	$\lambda = 3.284 + 1.617 \times 10^{-2} \cdot T - 2.305 \times 10^{-6} \cdot T^2$	403-1 100	(15%)

15.3.2 *Pipe wall friction loss correlations*

For the calculation of the distributed pressure loss all codes utilise a relation of Eq. (15.1), where the friction factor is evaluated with different correlations. These correlations are reported in Table 15.3.2 where information is provided on the range of applicability and literature references.

Table 15.3.2: Formulas for calculation of friction loss coefficients

		Correlations	Applicable range	Reference
Laminar flow		$f = \frac{64}{Re}$ (Hagen-Poiseuille law)	Up to $Re \approx 2\,000$	White (1986)
Transient or turbulent flow	1	$f_{L,T} = \left(3.75 - \frac{8250}{Re}\right)\left(f_{T,3000} - f_{L,2200}\right) + f_{L,2200}$ $f_{L,2200}$: Laminar friction factor at Re = 2 200 $f_{T,3000}$: Turbulent friction factor at Re = 3 000	2 200 < Re < 3 000	Colerbrook (1939) Zigrang (1985)
		$\frac{1}{\sqrt{f_T}} = -2\log_{10}\left\{\frac{\varepsilon}{3.7D} + \frac{2.51}{Re}\left[1.14 - 2\log_{10}\left(\frac{\varepsilon}{D} - \frac{21.25}{Re^{0.9}}\right)\right]\right\}$	Re > 3 000	
	2	$\frac{1}{\sqrt{f}} = -2\log_{10}\left\{\frac{\varepsilon}{3.7D} + \frac{2.51}{Re\sqrt{f}}\right\}$ (Colebrook's interpolation formula)	Re > 2 000	Idelchik (1994)
	3	$f = 0.11\left(\frac{68}{Re} + \frac{\varepsilon}{D}\right)^{0.25}$ (Altshul's approximate formula)	Re > 2 090·$(\varepsilon/D)^{0.635}$	Idelchik (1994)
	4	$f = \frac{1}{\left[2\log_{10}\left(\frac{2.51}{Re\sqrt{f}}\right) + \frac{\varepsilon}{3.7D}\right]^2}$	Re > 2 090·$(\varepsilon/D)^{0.635}$	Idelchik (1994)
	5	$f = \frac{0.3164}{Re^{0.25}}$ (Blasius)	4 000 < Re < 10^5	Idelchik (1994)
	6	$f = 8\cdot\left[\left(\frac{8}{Re}\right)^{12} + \frac{1}{(A+B)^{1.5}}\right]^{1/12}$ $A = \left\{2.457\cdot\ln\left[\left(\frac{7}{Re}\right)^{0.9} + 0.27\cdot\left(\frac{\varepsilon}{D}\right)\right]\right\}^{16}$ $B = \left(\frac{37530}{Re}\right)^{16}$ (Churchill's formula)	All ranges	Idelchik (1994)
	7	$f = \left(1.8\log_{10}\frac{8.25}{\frac{56}{Re} + \frac{\varepsilon}{D}}\right)^{-2}$	For turbulent flow	Slissky (1983)

f, r, D, Re_D and ε are friction factor, radius of pipe, diameter of pipe, Reynolds number and absolute surface roughness, respectively.

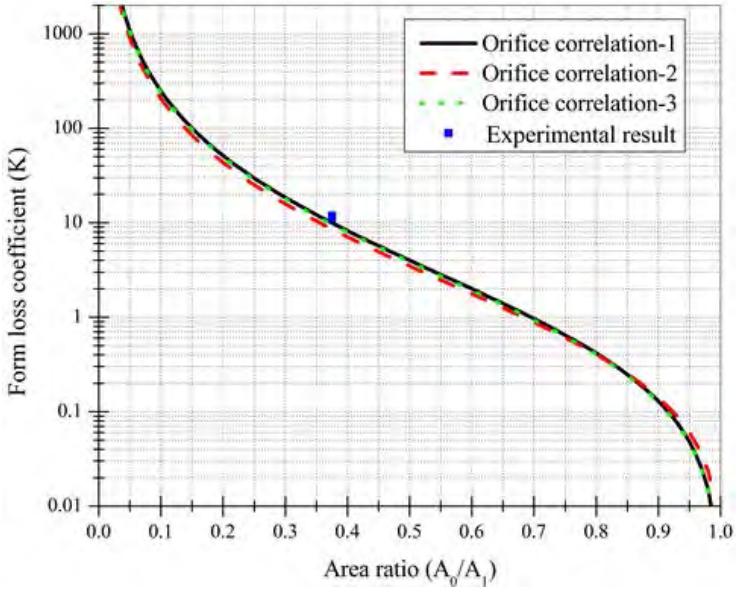
Table 15.3.3: Formulas for calculation of form loss coefficients in orifices

	Correlations	Applicable range	Reference
1	$K = \left(1 + 0.707\sqrt{1 - \frac{A_0}{A_1}} - \frac{A_0}{A_1}\right)^2 \left(\frac{A_1}{A_0}\right)^2$	$Re > 10^5$	Idelchik (1994)
2	$K = \left(1 - \frac{A_0}{A_1} + \sqrt{\xi}\left(1 - \frac{A_0}{A_1}\right)^{0.375}\right)^2 \left(\frac{A_1}{A_0}\right)^2$ where $\xi = 0.13 + 0.34 \times 10^{-\left[3.4(l/D) + 88.4(l/D)^{2.3}\right]}$	$Re > 10^4$	Idelchik (1994)
3	$K = \left(0.5\left(1 - \frac{A_0}{A_1}\right) + \left(1 - \frac{A_0}{A_1}\right)^2 + \tau\sqrt{1 - \frac{A_0}{A_1}}\left(1 - \frac{A_0}{A_1}\right)\right)\left(\frac{A_1}{A_0}\right)^2$ $\tau = (2.4 - \bar{l}) \times 10^{-\varphi(\bar{l})}$ $\varphi(\bar{l}) = 0.25 + \frac{0.535\bar{l}^{-8}}{0.05 + \bar{l}^{-8}}$	$Re > 10^3$	Idelchik (1994)

Where *f, r, D, Re_D* and ε are friction factor, radius of pipe, diameter of pipe, Reynolds number and absolute surface roughness, respectively.

All the correlations used in prediction of pressure loss for an orifice depend on the ratio of pipe flow area to orifice tap small area, regardless of Reynolds number, as shown in Figure 15.3.3. The area ratio (A_0/A_1) in the orifice of HELIOS is about 3.8. Experimental results (having a range of 10.9-11.6, almost constant independent of Reynolds numbers) show good agreement with used correlations.

Figure 15.3.3: Comparison of orifice correlations with experimental result



15.3.4 *Spacers*

Pressure losses of core or heat exchanger are dominant among those of the fluid system because of its small flow area and complex geometry. Grid spacers that support the core rods or steam generator tubes are the highest regions of pressure losses. Rehme (1973)

In contrast the detailed modelling of the complex geometry using CFD tools produced results that agree well with HELIOS experimental data, as shown in the figure above. It has been found that CFD modelling always provided more accurate results than experimental correlations (Cho, 2011). Further examples are given in Section 15.3.6.

15.3.5 *Sudden area change*

In regions like the gate valve and connecting piece of flange section, flow areas suddenly change, which disturbs the flow. The sudden area changes are illustrated in Figure 15.3.6 and correlations are listed in Table 15.3.5.

Figure 15.3.6: Geometries of sudden area changes

Sudden expansion (left), sudden contraction (right)

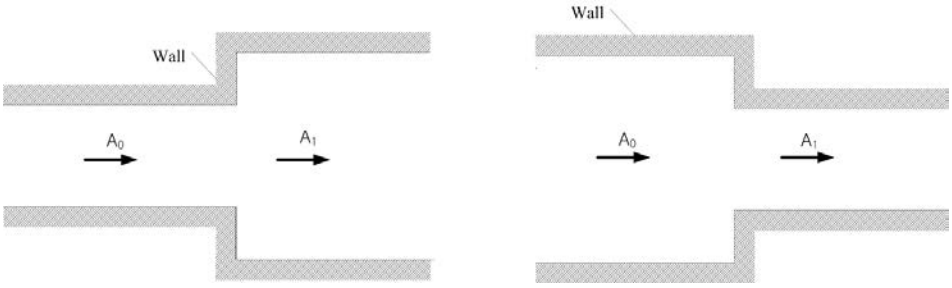


Table 15.3.5: Formulas for calculation of form loss coefficients in sudden expansion of flow area

Sudden expansion	Range	Reference
$K = \left(1 - \frac{A_0}{A_1}\right)^2$ Borda-Carnot's formula	$Re > 3.3 \cdot 10^3$	Idelchik (2000)

Table 15.3.6: Formulas for calculation of form loss coefficient in sudden contraction of flow area

	Sudden contraction	Range	Reference
1	$K = 0.5\left(1 - \frac{A_1}{A_0}\right)$	$Re > 10^4$	Idelchik (2000)
2	$K = 0.5\left(1 - \frac{A_1}{A_0}\right)^{3/4}$	$Re > 10^4$	Idelchik (1992)
3	$K = 0.5 - 0.7 \cdot \left(\frac{A_1}{A_0}\right) + 0.2 \cdot \left(\frac{A_1}{A_0}\right)^2$	$Re > 10^4$	Massey (1968)
4	$K = \left(1 - \frac{A_1}{A_c}\right)^2$ $\frac{A_c}{A_1} = 0.62 + 0.38\left(\frac{A_1}{A_0}\right)^3$	All range	Vennard (1961)